

Computer Networks



Overview of the Internet Protocol layering (Book 1 Ch 1)

Application Layer

- Application-Layer Paradigms
- Client-server applications
 - World Wide Web and HTTP
 - FTP
 - Electronic Mail
 - DNS
- Peer-to-peer paradigm
 - P2P Networks
- Case study: BitTorrent (Book 1 Ch 2)

① Introduction

- Providing Services
- Application-Layer Paradigms

② Standard Client-Server Applications

- World Wide Web and HTTP
- FTP
- Electronic Mail
- Domain Name System (DNS)

③ Peer-to-Peer Paradigm

- P2P Networks

Peer-to-Peer Paradigm

- Peer-to-peer (P2P):
 - Alternative to client-server paradigm
- In P2P:
 - Peers communicate directly
- Each peer can act as:
 - Client
 - Server
- P2P mainly used for:
 - File sharing
 - Distributed communication

Early Development of P2P

- First P2P file sharing system:
 - WWIVnet
- Developed by:
 - Wayne Bell
- Year:
 - December 1987
- Freenet introduced in:
 - July 1999
- Designed by:
 - Ian Clarke
- Features of Freenet:
 - Decentralized data store
 - Censorship resistant
 - Strong anonymity protection



Growth of Peer-to-Peer Applications

- Napster popularized P2P:
 - 1999–2001
- Napster created by:
 - Shawn Fanning
- Main purpose:
 - Online music file sharing
- Problem:
 - Copyright violation lawsuits
- Result:
 - Service closed
- Napster influenced later P2P systems:
 - Gnutella
 - FastTrack
 - BitTorrent
 - WinMX
 - GUNet

- Internet users willing to share resources:
 - Become peers
- Peers together form:
 - Peer-to-peer network
- Shared resources may include:
 - Audio files
 - Video files
 - Other data files
- Peer with a file:
 - Makes file available to others

File Sharing in P2P

- Interested peer:
 - Connects to source peer
- File downloaded directly from:
 - Another peer's computer
- After downloading:
 - Peer can also share same file
- As more peers join:
 - More copies of file become available
- Result:
 - Improved file distribution

Types of P2P Networks

- Number of peers in network:
 - Continuously changes
- Challenges in P2P:
 - Tracking active peers
 - Finding file locations
- P2P networks divided into:
 - Centralized P2P networks
 - Decentralized P2P networks
- Classification based on:
 - Method of managing peers and files

Centralized P2P Networks

- Centralized P2P network:
 - Uses both client-server and peer-to-peer paradigms
- Directory system uses:
 - Client-server model
- File storage and downloading use:
 - Peer-to-peer model
- Also called:
 - Hybrid P2P network
- Example:
 - Napster

Working of Centralized P2P

- Peer first:
 - Registers with central server
- Peer provides:
 - IP address
 - List of shared files
- User searching for file:
 - Sends query to central server
- Central server:
 - Searches directory
 - Returns IP addresses of peers having file
- Download process:
 - Peer downloads file directly from another peer
- Directory continuously updated:
 - As peers join or leave network

Advantages and Drawbacks

- Advantages:

- Easy directory management
- Faster file searching

- Drawbacks:

- Heavy traffic at central server
- System slowdown possible
- Central server vulnerable to attacks

- If central servers fail:

- Entire system becomes unavailable

- Napster issue:

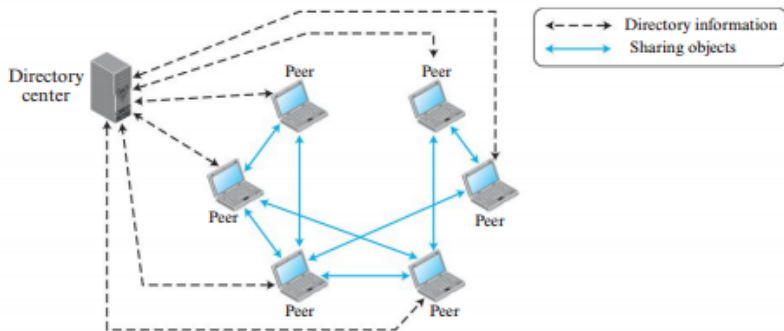
- Copyright lawsuits led to closure in 2001

- New Napster:

- Relaunched by Roxio in 2003
- Legal paid music service

Centralized network

Figure 2.46 Centralized network



Decentralized P2P Networks

- Decentralized P2P network:
 - Does not use central directory server
- Peers organize themselves:
 - Directly with other peers
- Network formed is called:
 - Overlay network
- Overlay network:
 - Logical network built over physical network
- File searching and sharing:
 - Performed without centralized control

Types of Decentralized P2P Networks

- Classification based on:
 - How peers are connected in overlay network
- Two types:
 - Unstructured P2P network
 - Structured P2P network
- Unstructured network:
 - Peers connected randomly
- Structured network:
 - Peers connected using predefined rules
- Decentralized networks provide:
 - Better scalability
 - No single point of failure

Unstructured P2P Networks

- In unstructured P2P networks:
 - Nodes are connected randomly
- Searching for a file:
 - Requires flooding queries through network
- Drawbacks:
 - High traffic overhead
 - Inefficient searching
 - Queries may fail
- Examples:
 - Gnutella
 - Freenet

Gnutella Network

- Gnutella:
 - Decentralized and unstructured P2P network
- Directory information:
 - Randomly distributed among nodes
- File search process:
 - Node A sends query to neighbor node W
- Query contains:
 - Identity of required object or file
- If node W knows location:
 - Sends address of node X
- Node A then:
 - Downloads file from node X

Flooding and Scalability in Gnutella

- If node W does not know file location:
 - Query flooded to all neighbors
- Flooding causes:
 - Heavy network traffic
- Main drawback:
 - Poor scalability
- Bootstrap process:
 - New node gets list of neighbor nodes
- Ping and pong messages used to:
 - Check whether neighbors are active

Improved Gnutella and Structured Networks

- Improved Gnutella introduced:
 - Ultra nodes
 - Leaf nodes
- Leaf nodes:
 - Do not perform routing
- Ultra nodes:
 - Handle routing and query propagation
- Additional techniques:
 - Query Routing Protocol (QRP)
 - Dynamic Querying (DQ)
- Structured networks:
 - Use predefined connection rules
 - Improve search efficiency
- Common structured technique:
 - Distributed Hash Table (DHT)
- Example:
 - BitTorrent